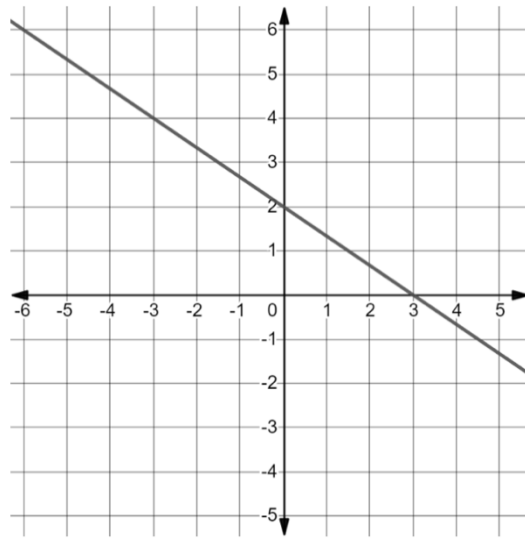


The graph of the function  $P(x)$  is shown.

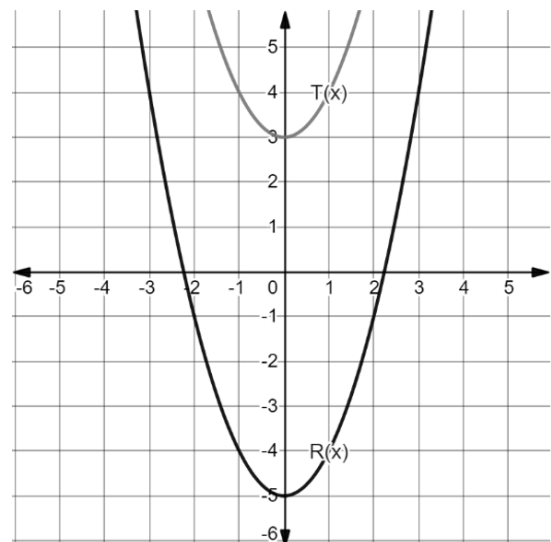


1. Write the function,  $P(x)$ , that represents the line.
2. Draw a line that is parallel to  $P(x)$  with a  $y$ -intercept of  $-2$ . Label this line  $Q(x)$ .
3. Describe how  $P(x)$  can be shifted to coincide with  $Q(x)$ .
4. Write the function  $Q(x)$  in terms of  $P(x)$ .

5. The graphs of  $R(x)$  and  $T(x)$  are shown on the graph.

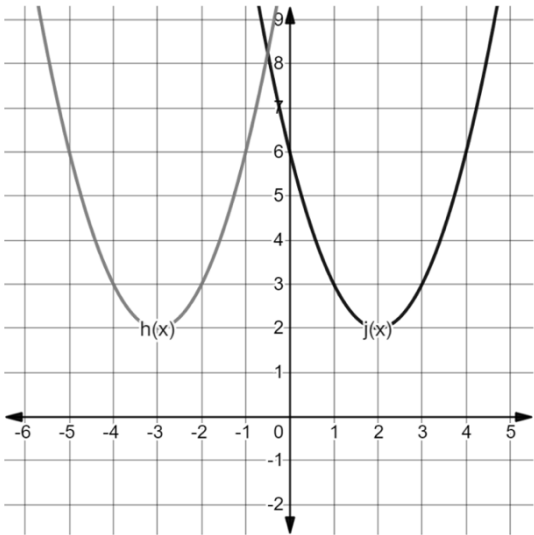
Write the transformation by completing the equation.

$$T(x) = R(x) + \underline{\hspace{2cm}}$$

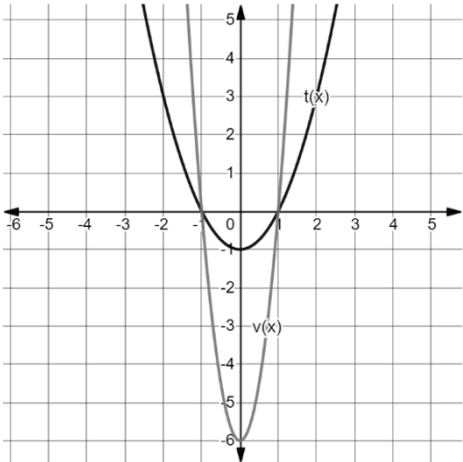


The functions  $h(x)$  and  $j(x)$  are shown on the graph.

6. Write  $h(x)$  in vertex form.
  
  
  
  
  
7. Write  $j(x)$  in vertex form.
  
  
  
  
  
8. Describe how  $h(x)$  was transformed to create  $j(x)$ .



9. The functions  $t(x)$  and  $v(x)$  are shown on the graph, where  $v(x)$  is a transformation of  $t(x)$ . What is the value of  $k$  such that  $v(x) = kt(x)$ ?



10. The table of values for  $w(x)$  and  $z(x)$  are shown, where  $z(x)$  is a transformation of  $w(x)$ . Write a description of the transformation.

| $y = w(x)$ |     | $y = z(x)$ |     |
|------------|-----|------------|-----|
| $x$        | $y$ | $x$        | $y$ |
| -2         | 8   | -2         | 13  |
| -1         | 2   | -1         | 7   |
| 0          | 0   | 0          | 5   |
| 1          | 2   | 1          | 7   |
| 2          | 8   | 2          | 13  |